

Trainings ▾

General Specifications

General Information

Listed below are general specifications for the KTA50, QSK50 and QSK50 CM850 engines. Additional specifications can be found in each system section.

Engine Type

KTA50	4 cycle, 60 degree vee, 16 cylinder
QSK50	4 cycle, 60 degree vee, 16 cylinder
Bore and Stroke	159 mm [6.25 in] x 159 mm [6.25 in]
Displacement	50.3 liters [3068 in ³]
Aspiration	Turbocharged and Aftercooled

Engine Weight (dry, estimated):

KTA50-C	4853 kg [10,700 lb]
KTA50-G	5361 kg [11,820 lb]
KTA50-M	5431 kg [11,973 lb]
QSK50, Generator Set Single-Stage with electronically actuated injectors	5620 kg [12,390 lb]
QSK50, Industrial Two-Stage with electronically actuated injectors	6050 kg [13,338 lb]
Crankshaft Rotation (viewed from the front of the engine)	Clockwise

Firing Order

Pre-September 1986	1R-1L-3R-3L-7R-7L-5R-5L-8R-8L-6R-6L-2R-2L-4R-4L
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After September 1986	1R-1L-3R-3L-2R-2L-5R-4L-8R-8L-6R-6L-7R-7L-4R-5L
QSK50 Electronic Reference	2-1-6-5-4-3-10-7-16-15-12-11-14-13-8-9
Compression Ratio	
with mechanically actuated injectors	13.5:1 to 15.5:1
with electronically actuated injectors (all applications)	15.0:1
Valve and Injector Settings:	
Exhaust	0.69 mm [0.027 in]
Intake	0.36 mm [0.014 in]
Maximum Overspeed Capacity	2150 rpm
Maximum allowable installed engine power angle	6 degree
Maximum allowable installed engine tilt angle	6 degree
Listed below are general specifications for the QSK38 engines. Additional specifications can be found in each system section.	
Engine Type	
QSK38	4 cycle, 60 degree vee, 12 cylinder
Bore and Stroke	159 mm [6.25 in] x 159 mm [6.25 in]
Displacement	37.7 liters [2301 in ³]
Aspiration	Turbocharged and Aftercooled
Crankshaft Rotation (viewed from the front of the engine)	Clockwise
Firing Order QSK38 Electronic Reference	2-11-10-3-6-7-12-1-4-9-8-5
Compression Ratio:	13.5:1 to 15.5:1
Valve and Injector Settings:	
Exhaust	0.69 mm [0.027 in]
Intake	0.36 mm [0.014 in]

Listed below are general specifications for the KTA38 and QSK38 engines. Additional specifications can be found in each system section.

Engine Type

KTA38	4 cycle, 60 degree vee, 12 cylinder
QSK38	4 cycle, 60 degree vee, 12 cylinder
Bore and Stroke	159 mm [6.25 in] x 159 mm [6.25 in]
Displacement	37.7 liters [2301 in ³]
Aspiration	Turbocharged and Aftercooled

Engine Weight (dry, estimated):

KTA38-C	3719 kg [8199 lb]
KTA38-G	4201 kg [9261 lb]
KTA38-M	4218 kg [9299 lb]
KTA38	4510 kg [9943 lb]
Crankshaft Rotation (viewed from the front of the engine)	Clockwise
Firing Order KTA38	1R-6L-5R-2L-3R-4L-6R-1L-2R-5L-4R-3L
Firing Order QSK38 Electronic Reference	2-11-10-3-6-7-12-1-4-9-8-5
Compression Ratio:	13.5:1 to 15.5:1

Valve and Injector Settings:

Exhaust	0.69 mm [0.027 in]
Intake	0.36 mm [0.014 in]

Listed below are general specifications for the KTA50, QSK50 and QSK50 CM850 engines. Additional specifications can be found in each system section.

Engine Type

KTA50	4 cycle, 60 degree vee, 16 cylinder
QSK50	4 cycle, 60 degree vee, 16 cylinder

Bore and Stroke	159 mm [6.25 in] x 159 mm [6.25 in]
Displacement	50.3 liters [3068 in ³]
Aspiration	Turbocharged and Aftercooled
Engine Weight (dry, estimated):	
KTA50-C	4853 kg [10,700 lb]
KTA50-G	5361 kg [11,820 lb]
KTA50-M	5431 kg [11,973 lb]
QSK50, Generator Set Single-Stage with electronically actuated injectors	5620 kg [12,390 lb]
QSK50, Industrial Two-Stage with electronically actuated injectors	6050 kg [13,338 lb]
Crankshaft Rotation (viewed from the front of the engine)	Clockwise
Firing Order	
Pre-September 1986	1R-1L-3R-3L-7R-7L-5R-5L-8R-8L-6R-6L-2R-2L-4R-4L
After September 1986	1R-1L-3R-3L-2R-2L-5R-4L-8R-8L-6R-6L-7R-7L-4R-5L
QSK50 Electronic Reference	2-1-6-5-4-3-10-7-16-15-12-11-14-13-8-9
Compression Ratio	
with mechanically actuated injectors	13.5:1 to 15.5:1
with electronically actuated injectors (all applications)	15.0:1
Valve and Injector Settings:	
Exhaust	0.69 mm [0.027 in]
Intake	0.36 mm [0.014 in]
Maximum Overspeed Capacity	2150 rpm
Maximum allowable installed engine power angle	6 degree

Maximum allowable installed engine tilt angle

6 degree

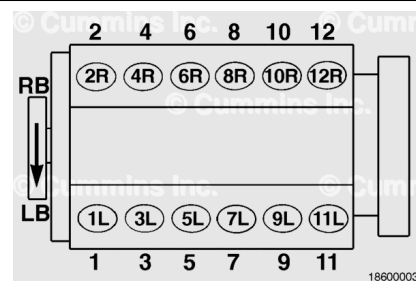
Note : Injector settings are not required for engines with electronically actuated injectors.

This graphic illustrates the cylinder numbering sequence for the KTA38 and QSK38 engines.

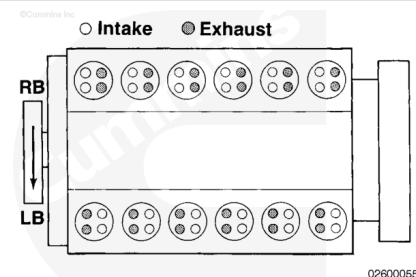
- RB = Right-bank of cylinders
- LB = Left-bank of cylinders.

To determine the right and left bank on the KTA38 and QSK38 engines, stand at the rear of the engine and face the front.

When referencing system electronics, a standard cylinder numbering system has been adopted. This is only used for the electronic system components on engines with electronically actuated injectors.



This graphic illustrates the intake and exhaust valve locations for the KTA38 and QSK38 engines.

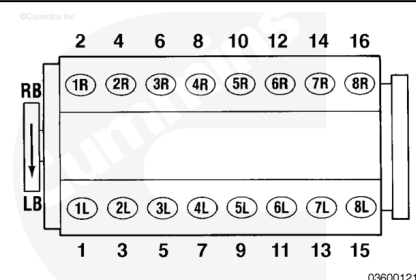


This graphic illustrates the cylinder numbering sequence for the KTA50 and QSK50 engines.

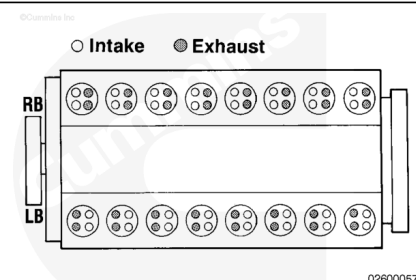
- RB = Right-bank of cylinders
- LB = Left-bank of cylinders.

To determine the right and left bank on the KTA50 and QSK50 engines, stand at the rear of the engine and face the front.

When referencing system electronics, a standard cylinder numbering system has been adopted. This is **only** used for the electronic system components on engines with electronically actuated injectors.



This graphic illustrates the intake and exhaust valve locations for the KTA50 and QSK50 engines.



Last Modified: 03-May-2019